When Directional Accuracy Lies: A Base-Rate-Honest Benchmark for LoRA-Adapted TimesFM on Equity Forecasting

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Abstract

Large pretrained time-series models such as TimesFM are attractive starting points for financial forecasting, but *raw directional accuracy is a misleading scoreboard in equity markets.* An early LoRA adapter in this project appeared to reach roughly 80% directional accuracy; we show this is not evidence of forecasting skill. Over a long horizon in a rising market, a trivial "always-up" rule attains comparably high accuracy *without using the input at all.* To separate genuine skill from this base-rate artifact, we build a reproducible, frozen-data benchmark with expanding *walk-forward* train/validation/test folds, a stratified *held-out-ticker* split, honest baselines (zero-shot TimesFM, always-up, random-walk, persistence, AR(1)), and paired significance tests (McNemar, Diebold–Mariano) under Benjamini–Hochberg FDR control. We apply the *identical* method to two universes—a tech-heavy NASDAQ-100 and a broad S&P 500—and report the headline metric as **excess accuracy over the always-up base rate**. Three findings emerge and replicate. First, when the historical ∼80% condition is recreated (a 2014+ bull window scored with raw accuracy), the high number is a base rate of ∼0.70, and the fine-tuned model scores *below* it. Second, on the honest benchmark, pooled LoRA shows **no directional skill over the base rate** at any horizon on either universe (excess accuracy centered on zero, negative at the six-month horizon). Third, on the pre-registered confirmatory test, **per-sector specialization is significantly *worse* than a single pooled adapter** (Diebold–Mariano $p < 0.001$ on held-out stocks at $h = 128$). Fine-tuning's only measurable benefit is a statistically significant reduction in point-forecast error relative to zero-shot TimesFM, which nonetheless does not beat naive baselines and confers no tradeable directional edge. The contribution is methodological: a defensible, fully seeded protocol that prevents the base-rate trap, together with the replicated negative result it produces.

1 Introduction

Financial forecasting is an unusually treacherous setting for large time-series models, because small apparent gains are easily confused with market drift, data leakage, or evaluation artifacts. The problem is most acute for *directional* accuracy. A model that predicts "up" over a six-month equity horizon can look highly accurate during a bull market even if it has learned nothing about the relationship between past and future prices.

This project began with an apparent regression: a first TimesFM LoRA adapter reported roughly 80% directional accuracy, while later per-sector adapters dropped toward 50%. Rather than treat the lower number as a failure, we treat the original 80% as the *suspect* result. The flaw is that the early evaluation compared raw accuracy against an implicit 50% coin flip, even though long-horizon equity direction is far from balanced: if most windows in the test period move up, an always-up rule scores very well with zero skill.

The central question is therefore not "can the model reach 80%?" but *"does the model beat the right baseline on the same windows?"* We design and implement a benchmark to answer this defensibly, and we run it on two universes with the method held fixed, so that the conclusion does not hinge on a single dataset. Our contributions are:

1. A frozen, checksum-versioned dataset and a fully **seeded, reproducible** training+evaluation pipeline: every reported number traces to a committed `raw.json` and a `run_meta.json` recording seed, code commit, clean-tree flag, and environment.

2. A walk-forward train/validation/test protocol that never early-stops on the test window, plus a stratified held-out-ticker split for symbol generalization.

3. The headline metric `excess_acc` = model accuracy − `always_up` accuracy, reported with block-bootstrap intervals and paired significance tests.

4. A three-method, two-universe study (legacy pooled, walk-forward pooled, and per-sector adapters; NASDAQ-100 and S&P 500) showing the negative result is robust and resolving the pre-registered per-sector-versus-pooled hypothesis.

2 Background and Related Work

TimesFM [1] is a patched, decoder-only time-series foundation model with internal reversible instance normalization (RevIN) [3] and a quantile output head. We adapt `google/timesfm-2.5-200m-pytorch` with LoRA [2], which inserts a small number of trainable low-rank matrices into selected linear layers while freezing the pretrained weights, making it cheap to train specialized adapters.

The closest motivation is the financial fine-tuning study of Fu, Hirano, and Imajo [4], which reports a *modest* directional edge ($\approx$ 54% at short horizons) rather than extreme accuracy. That reference is exactly why an 80% jump should not be accepted without checking base-rate baselines, leakage, and out-of-sample generalization. We also draw on standard forecast-evaluation practice: walk-forward validation, random-walk and persistence baselines, MASE for point error [5], quantile coverage and pinball loss [6], Diebold–Mariano tests [7, 8], McNemar's paired test [9], the block bootstrap [10], and Benjamini–Hochberg FDR control [11].

3 Research Questions and Pre-registration

RQ1 (primary, confirmatory). Was the apparent directional accuracy a base-rate artifact? Tested by `excess_acc` = acc − `always_up`-acc on identical windows.

RQ2 (confirmatory). Does per-sector specialization improve on a single pooled adapter? Pre-registered as *per-sector vs. pooled on held-out S&P 500 stocks at* $h = 128$ (Diebold–Mariano).

RQ3. Does LoRA fine-tuning improve on zero-shot TimesFM?

RQ4 (exploratory). Does the conclusion depend on the universe? We replicate the identical method on a tech-heavy NASDAQ-100 and a broad S&P 500.

Honesty note. The per-sector test (RQ2) is reported on S&P 500 only; NASDAQ-100 sector strata are too thin to train per-sector adapters (e.g. a single financials name), so the universe comparison (RQ4) is restricted to the pooled adapter and is treated as *exploratory*, not as a registered hypothesis.

4 Data

Each universe is frozen once into a versioned, checksum-verified artifact and reused by every run, so results never depend on a re-download (Table 1). Price field is the split/dividend-adjusted close. The NASDAQ-100 membership and sector tags are sourced from the advisor's `simofi` catalog and mapped onto a single sector taxonomy shared with the S&P 500.

Table 1: Frozen datasets (2005-01-01 to 2026-01-01, daily).

	NASDAQ-100	S&P 500
Stocks (+ index/ETFs)	100 (+QQQ)	501 (+11 SPDR ETFs)
Sectors	10 (tech-heavy: 41 tech)	11 (balanced)
Seen / held-out stocks	76 / 24	394 / 107
Sectors with a held-out set	7 / 10	11 / 11

Limitation (survivorship). Both universes are *current-membership snapshots*, not point-in-time constituents. This inflates the up base rate and the long-horizon upward drift, so results are a benchmark over a frozen current-constituent universe, not a live investable simulation (Section 8).

5 Methodology

5.1 Walk-forward folds

Evaluation uses three expanding folds; validation is the *only* signal for early stopping and model selection, and the test window is never used for selection (Table 2).

Table 2: Expanding walk-forward folds (train → validation → test).

Fold	Train	Validation	Test
0	2005–2019	2019–2020	2020–2022
1	2005–2021	2021–2022	2022–2024
2	2005–2023	2023–2024	2024–2026

5.2 Held-out tickers

Within each sector, stocks are split ∼80/20 into *seen* and *held-out* with a fixed seed (42), *fixed across folds*, so held-out names are never trained in any fold. ETFs are excluded from the holdout pool and reported separately. Every method is scored on both splits.

5.3 Normalization (leakage control)

A per-series log transform followed by z-scoring is fit only on *pre-target history*: the train period for validation windows, train+validation for test windows. No target-window observation is used. Direction is invariant to this monotone transform; returns and point error use inverse-transformed prices.

5.4 Reproducibility

Training is fully seeded (seed = 42 across Python/NumPy/torch/CUDA, with a seeded DataLoader generator; the seed is set *before* LoRA initialization). The adapter scored is the *best-validation* checkpoint, not the last epoch. Each run writes a `run_meta.json` manifest recording seed, git commit, a tracked-files-only clean-tree flag, GPU, library versions, the full dependency freeze, hyperparameters, and the per-adapter best epoch. All runs in this paper were produced at one code commit on a clean tree, seed 42, on an NVIDIA A100 (torch 2.11.0+cu128, Python 3.12); a two-run reproducibility gate confirmed bit-identical Table-A accuracies. We do not enable fully deterministic CUDA kernels (they raise on TimesFM operators), so results are single-seed deterministic up to negligible GPU-atomic jitter; multi-seed robustness is left to future work.

5.5 Models and baselines

Zero-shot TimesFM is the pretrained base with no adapter, served from a dedicated instance with an assertion that adapter loading cannot mutate it. **Pooled LoRA** is one adapter trained on all seen stocks. **Per-sector LoRA** trains one adapter per GICS sector (seen stocks of that sector plus its SPDR ETF anchor) and routes each ticker to its sector's adapter at evaluation. Naive baselines are `always_up` (predict up every window—the base rate), random-walk (last value; no directional call), persistence (repeat last return sign), and AR(1) (direct h-period return). All adapters share the configuration in Table 3; only the data differ across universes and methods.

Table 3: Training configuration (identical across universes and methods).

Base	timesfm-2.5-200m	LoRA rank / α / dropout	32 / 64 / 0.05
Context / horizon	512 / 128	Batch	512
Optimizer	AdamW (lr 10^{-4}, wd 0.01)	Schedule	cosine warm restarts
Grad clip	1.0	Loss	MSE $+ 0.3 \cdot$directional
Max epochs / patience / min	80 / 15 / 30	Seed	42

5.6 Metrics

The headline directional metric is `excess_acc` = acc − `always_up`-acc on identical windows; we also report balanced accuracy and Matthews correlation. Point error (price space, per-ticker then macro-averaged) uses MAE, RMSE, sMAPE, and MASE. Calibration uses empirical quantile coverage versus nominal and pinball loss. Economic metrics use real per-trade returns on non-overlapping h-day windows; maximum drawdown is computed on a unit-stake equity curve with per-trade returns floored at -100% (a margin stop), so a short squeeze cannot drive it below -1.

5.7 Significance

Excess accuracy versus always-up uses the paired McNemar test plus a block-bootstrap interval; forecast-error comparisons use Diebold–Mariano on normalized-space errors with a sample-size-capped Newey–West variance (negative $\Rightarrow$ first method has lower loss). The exploratory family is FDR-controlled by Benjamini–Hochberg.

6 Results

All headline numbers are on *held-out stocks*, averaged over the three walk-forward folds. Figures are produced directly from the committed result files.

6.1 Recreating the legacy ∼80% as a base-rate artifact (RQ1)

We first recreate the historical condition that produced the ∼80% claim: a pooled adapter trained on a 2014+ bull window (train 2014–2023, early-stop 2023–2024) and scored with raw accuracy on the 2024–2026 test window. The recreated base rate is high but the model lands *below* it (Table 4): at $h = 128$ the always-up rule alone scores 0.704 on held-out stocks, while the fine-tuned model scores 0.626 (`excess_acc` = −0.08). The original 0.778 figure was computed on only ∼45 windows and does not reproduce on the full universe; honestly measured, the "high accuracy" is a base rate of ∼0.70 that the model does not beat. The mechanism behind the 80% claim—high raw accuracy that looks impressive against a 50% reference—is therefore confirmed, and shown to be the trend rather than skill.

Table 4: Legacy 2014+ condition, held-out stocks: the recreated base rate is high, but the model scores below it (no excess skill).

Horizon h	8	32	64	128
`always_up` base rate	0.684	0.629	0.636	0.704
Model raw accuracy	0.559	0.526	0.530	0.626
Excess (model − base rate)	−0.125	−0.103	−0.106	−0.078

6.2 No directional skill over the base rate (RQ1, both universes)

On the honest walk-forward benchmark, raw accuracy is high at the long horizon (0.63–0.71) but tracks the always-up base rate; excess accuracy is centered on zero with bootstrap intervals spanning zero at every horizon, and is negative at $h = 128$ (Table 5, Figure 1). Zero-shot TimesFM is *below* the base rate everywhere. At $h = 128$ the base rate alone is 0.710 (NASDAQ) and 0.658 (S&P), above pooled accuracy of 0.630 and 0.641: no configuration beats the base rate at the long horizon on either universe.

Table 5: Excess accuracy (model − `always_up`), held-out stocks, by horizon. Positive means skill over the base rate; intervals span zero throughout.

h	2	4	8	16	32	64	128
NASDAQ-100							
pooled	+0.036	+0.061	−0.024	+0.006	+0.001	−0.000	−0.081
zero-shot	−0.047	−0.007	−0.053	−0.017	−0.062	−0.011	−0.151
S&P 500							
pooled	+0.004	−0.009	+0.002	+0.019	−0.005	−0.017	−0.017
per-sector	−0.031	−0.036	−0.039	+0.001	−0.045	−0.044	−0.059
zero-shot	−0.024	−0.022	−0.074	−0.028	−0.080	−0.059	−0.125

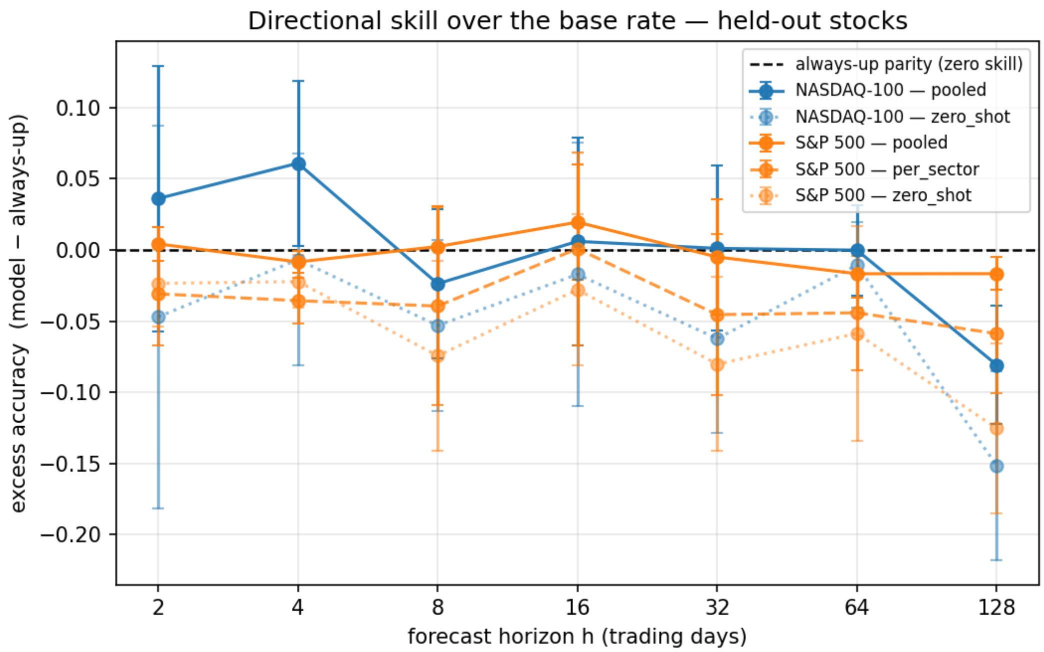

Figure 1: Excess directional accuracy versus horizon, held-out stocks, both universes. The dashed line is always-up parity (zero skill). Pooled curves hug the line with intervals spanning zero; zero-shot sits consistently below it.

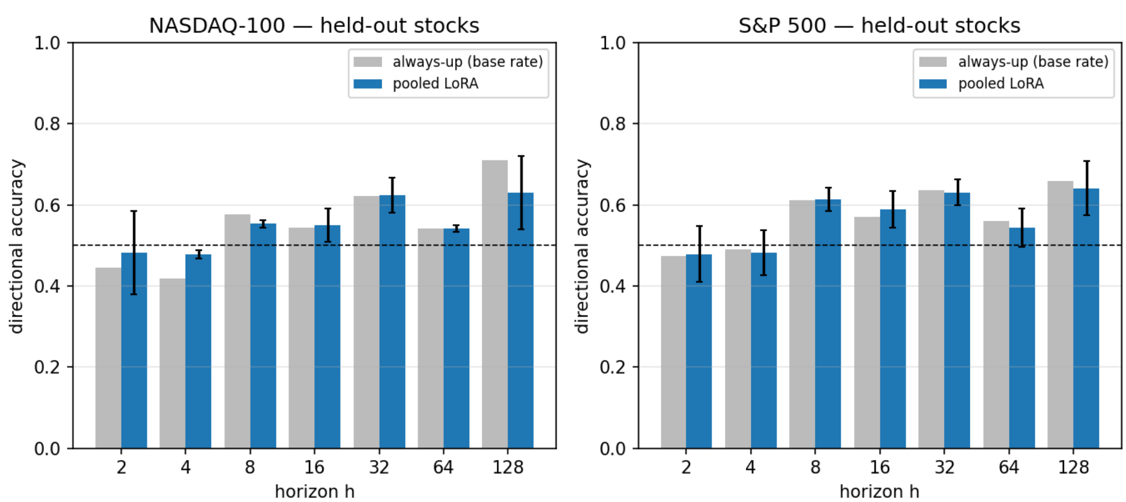

Figure 2: Raw directional accuracy of pooled LoRA tracks the always-up base rate at every horizon; at $h = 128$ the base rate is higher. High raw accuracy is not skill.

6.3 Per-sector versus pooled: the pre-registered test (RQ2)

The registered confirmatory comparison resolves cleanly against specialization. On held-out S&P 500 stocks, per-sector accuracy (0.599 at $h = 128$) is below pooled (0.641) and below the base rate (0.658), and the Diebold–Mariano test rejects equality *in pooling's favor*: per-sector has significantly higher forecast loss at the long horizon (Table 6). Sector specialization therefore *hurts* relative to a single pooled adapter—most plausibly because each sector adapter trains on far less data—and neither beats the base rate.

Table 6: Pre-registered test: per-sector vs. pooled, held-out S&P 500 stocks, Diebold–Mariano statistic per fold (positive $\Rightarrow$ per-sector has higher loss, i.e. *worse*). Significant in pooling's favor at the long horizon.

Horizon	Fold 0	Fold 1	Fold 2
$h = 8$	+3.26 (p=.001)	+2.15 (p=.031)	+3.57 (p<.001)
$h = 32$	+3.43 (p=.001)	+1.28 (p=.20)	+2.20 (p=.028)
$h = 128$	+5.12 (p<.001)	+1.03 (p=.30)	+5.50 (p<.001)

6.4 Fine-tuning versus zero-shot: point error only (RQ3)

Pooled LoRA has *lower* point error than zero-shot TimesFM on both universes (held-out MAE 15.15 vs. 16.31 on S&P; 25.15 vs. 26.44 on NASDAQ; Table 7, Figure 3), and Diebold–Mariano rejects equality in pooled's favor in a majority of cells after FDR (15/21 S&P-seen, 12/21 S&P-held-out; 7/21 and 4/21 on NASDAQ). However, pooled does *not* beat the naive always-up point forecast (15.56 on S&P, 24.91 on NASDAQ), and the directional and economic edge is absent. Fine-tuning sharpens the point forecast without conferring tradeable skill—consistent with the observation that the best-validation epoch is reached within a few epochs (S&P: $\{1, 2, 1\}$; NASDAQ: $\{5, 16, 3\}$ of 44 run), after which validation loss rises.

Table 7: Point-forecast error (MAE, held-out stocks). LoRA < zero-shot, but ties the naive baseline. MAE is not comparable across universes (different price levels).

	pooled	zero-shot	`always_up`
NASDAQ-100	25.15	26.44	24.91
S&P 500	15.15	16.31	15.56

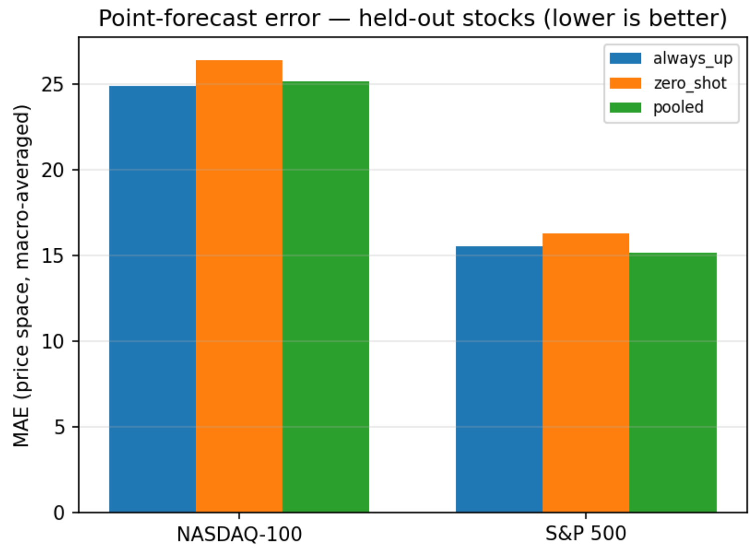

Figure 3: Point-forecast MAE on held-out stocks: pooled LoRA is below zero-shot but ties the always-up baseline.

6.5 Calibration and statistical power

The quantile tails are poorly calibrated for both models on both universes: the nominal-0.05 quantile has ∼0.44 empirical coverage (pooled 0.440 / zero-shot 0.408 on S&P; 0.436 / 0.404 on NASDAQ), a base-model trait LoRA does not fix (Figure 4). The long horizon is severely underpowered: at $h = 128$ the held-out McNemar discordant-pair count per fold is only $\{5, 5, 20\}$ (S&P) and $\{3, 10, 10\}$ (NASDAQ), so we make *no* significance claim at $h = 128$ for the directional tests; it is reported for completeness.

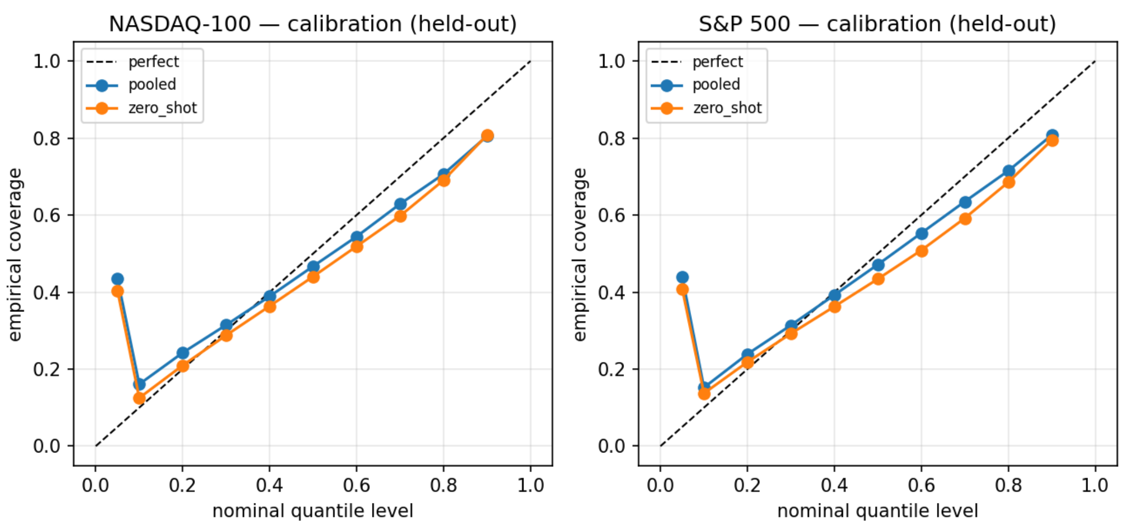

Figure 4: Quantile reliability (held-out stocks): empirical coverage versus nominal for pooled and zero-shot. Both are poorly calibrated in the tails on both universes.

6.6 Universe replication (RQ4)

The two universes differ in size (100 vs. ∼500 names), sector mix (41% vs. ∼15% technology), and price level, yet the qualitative result is identical: excess accuracy ≈ 0, negative at $h = 128$, zero-shot below the base rate, and LoRA better than zero-shot on point error only. We therefore do *not* claim one universe is "better" (the comparison is confounded); the value of the replication is that the negative finding is robust to the universe rather than an artifact of one dataset.

7 Discussion

The interpretive risk this project set out to avoid—mistaking raw directional accuracy for skill—is exactly the trap the original 80% fell into. With an honest base-rate baseline on identical windows, excess accuracy collapses to zero and is negative at the six-month horizon: the equity uptrend, not the model, produced the high accuracy, and this holds on a tech-heavy and a broad universe alike. The pre-registered comparison adds a second, sharper negative: sector specialization is significantly worse than pooling, so the intuitive "train a specialist per sector" recipe is counterproductive here, most plausibly because each adapter sees far less data.

The one genuine, statistically supported effect is that LoRA reduces point-forecast error relative to zero-shot TimesFM. This is a real but narrow benefit: it does not beat naive baselines, does not produce directional accuracy over the base rate, and does not yield a positive risk-adjusted return. Combined with the early best-validation epoch, the picture is of a base model already near its useful operating point on these series, with LoRA offering a small refinement of magnitude but not of direction. For practice the lesson is methodological: report excess-over-base-rate, hold out tickers, walk forward, and test paired significance before claiming a foundation model "works" on equities.

8 Limitations

1. **Long-horizon power.** Non-overlapping $h = 128$ windows yield only a few hundred (S&P) or few dozen (NASDAQ) windows per fold and $\{5, 5, 20\}/\{3, 10, 10\}$ McNemar discordant pairs; $h = 128$ directional significance is not estimable and is not claimed.

2. **Survivorship bias.** Current-membership snapshots inflate the up base rate and long-horizon drift; results do not transfer unchanged to a point-in-time universe.

3. **Universe comparison is confounded.** NASDAQ vs. S&P differ in size, sector mix, and price level, so no "which universe is better" claim is made.

4. **Single seed.** Results are single-seed deterministic, not multi-seed robust.

5. **Windowing.** Non-overlapping windows (stride = horizon) under-sample short horizons.

6. **Scope.** Equities and sector ETFs only; no forex or macro; CRPS is an even-grid quantile approximation.

Future work: multi-seed mean±SD; a matched 100-stock S&P subsample to deconfound universe size from sector mix; point-in-time membership to remove survivorship; and overlapping windows or longer test spans to recover long-horizon power.

9 Conclusion

Across two equity universes, with an identical seeded, reproducible method, LoRA-adapted TimesFM shows no directional skill over the always-up base rate at any horizon, zero-shot TimesFM is worse than the base rate, and per-sector specialization is significantly worse than a single pooled adapter. Fine-tuning's only measurable benefit is lower point-forecast error than zero-shot, which beats neither the naive baselines nor the base rate. The apparent ∼80% directional accuracy that motivated the project was a base-rate/trend artifact—recreated here as a ∼0.70 base rate the model fails to beat. The deliverable is a benchmark protocol that makes such artifacts visible, and the replicated, honest negative result it produces.

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